

Software Requirements Specification

Public Website and Content Management Portal

Phase 1 — Implementation Baseline

Document control	Value
Project	2D Montessori School Website and Admin Portal
Document version	1.0
Status	Approved Phase 1 scope — ready for estimation and implementation
Prepared for	2D Montessori (OPC) Private Limited
Prepared by	Project requirements consolidation
Date	12 August 2026
Default timezone	Asia/Kolkata
Primary language	English (Tamil-ready architecture)

Purpose

This SRS converts the approved product decisions into a testable baseline for design, development, quality assurance, deployment and acceptance. Any change affecting cost, schedule, data handling or permissions should be managed through formal change control.

1. Executive Summary

The product will provide a responsive, accessible and search-friendly public website for 2D Montessori, supported by a secure role-based administration portal. Phase 1 focuses on publishing school information, managing media and enquiries, and maintaining content through a controlled review workflow. It deliberately excludes full online admissions, payments, multilingual authoring and advanced automation.

Outcome	Phase 1 commitment
Public presence	Mobile-first website covering school, Montessori approach, programs, experiences, admissions, team, events, gallery, articles, FAQs, testimonials, documents and contact information.
Content operations	Structured admin modules with drafts, review comments, approval, preview, scheduling, version history, rollback, archive and recycle-bin controls.
Enquiry operations	Capture, assign, follow up, filter and export enquiries; reminders remain in-app.
Trust and quality	Role permissions, audit logs, consent metadata, legal pages, cookie controls, SEO, accessibility, performance and production checks.
Delivery boundary	Information and enquiry collection only; no applicant portal, fee payment, document upload or automated email/WhatsApp messaging.

2. Scope and Boundaries

2.1 In scope

- Responsive public website and controlled homepage, header, navigation and footer.
- Programs, Experiences, Montessori Way, Admissions, Team, Events/Announcements, Gallery, Testimonials, Blog/News, FAQs, downloadable documents and Contact.
- Admin authentication by email/password; Super Admin-created accounts; first-login password change.
- Role-based content lifecycle, preview, scheduled publishing, validation, autosave, version history and rollback.
- Media library with image optimisation, usage tracking and child-image consent metadata.
- Enquiry assignment, follow-up records, dashboard reminders and CSV/Excel export.
- Legal/policy pages, cookie consent, global search, SEO, URL redirects, analytics and activity logging.
- English-only Phase 1 with data structures prepared for future Tamil content.

2.2 Out of scope for Phase 1

- Full online admission applications, applicant tracking, child document uploads and fee payments.
- Google Sign-In, invitation emails, automated email/WhatsApp notifications and social auto-posting.
- Tamil authoring/translation interface and multilingual public pages.
- Shared categories/tags and automatic related-content recommendations.
- Advanced spam protection, conditional CAPTCHA, rate limiting and duplicate-submission intelligence beyond basic validation/security rules.
- CSV/Excel content import, advanced bulk workflow actions, concurrent edit locking, automatic video transcoding/thumbnail generation and advanced duplicate-media detection.
- Ratings, video testimonials and parent feedback collection forms.

2.3 Assumptions and dependencies

- The existing domain 2dmontessori.com and Firebase project remain available.

- Brand assets, approved copy, school address, contact details, policies and media consents will be supplied by authorised stakeholders.
- Google Maps, analytics and Search Console configuration require valid project ownership and production credentials.
- Final file-size, storage and video-delivery limits will be agreed before build completion.
- Any feature not explicitly listed in this SRS requires change approval.

3. Users, Roles and Permissions

Capability	Editor	Publisher	Super Admin
Create/edit own drafts	Yes	Yes	Yes
Edit all content	Assigned/permitted	Yes	Yes
Submit for review	Yes	Yes	Yes
Approve/reject/publish	No	Yes	Yes
Schedule publishing	Suggest only	Yes	Yes
Manage redirects/SEO	Limited content fields	Yes	Yes
Manage users and roles	No	No	Yes
Delete enquiries permanently	No	No	Yes
Export enquiries/content	No	No	Yes
Manage brand/system settings	No	No	Yes
View activity log	Own/relevant	Relevant	All

AUTH-01 Account creation

- Only the Super Admin creates user accounts using name, email, role and a temporary password. No invitation or notification email is sent in Phase 1.

AUTH-02 Authentication

- Users sign in using email and password. Passwords are handled by Firebase Authentication and must never be stored in Firestore or logs.

AUTH-03 First login and password management

- A newly created or reset account must change its temporary password at first login. Users can later change their password from My Profile.

AUTH-04 Profile management

- Users may edit full name and profile photograph. Email, role, status, audit fields and must-change-password status remain system-controlled.

AUTH-05 Administration

- Super Admin can change email/role, reset access with a temporary password, suspend/reactivate accounts and review login/account activity.

4. Public Website Requirements

WEB-01 Homepage

- Controlled sections: notice banner; hero with text, media and CTA; introduction; featured Programs; Montessori Way highlights; Experiences; events; gallery; testimonials; latest articles; admissions CTA; contact/location. Approved sections may be reordered or hidden.

WEB-02 Header and footer

- Central logo/brand treatment; responsive navigation; controlled dropdown depth; search; enquiry/admissions CTA; optional phone/WhatsApp shortcuts; useful/program/legal/social/contact links; copyright and Cookie Settings.

WEB-03 Core editorial pages

- About, Montessori Way and other approved pages support structured sections, media, calls to action, SEO, preview and publishing workflow.

WEB-04 Programs and Experiences

- Structured listing/detail pages, featured selection, media, FAQs, testimonials and manually selected related content.

WEB-05 Admissions

- Editable process, eligibility, ages, academic year, required documents, timings, visit/interaction details, FAQs, brochure, notices and Enquire/Book a Visit/Call/WhatsApp actions.

WEB-06 Events and Announcements

- Events, holidays, workshops, admission notices and urgent updates with dates, venue, media, links, feature/urgent flags, scheduling and expiry. Expired items are hidden publicly but retained in admin.

WEB-07 Gallery

- Published albums/images with captions, alt text and consent controls. Public media must use optimised responsive renditions.

WEB-08 Blog/News

- Listing and article pages with author/date, social sharing, SEO, related content, scheduled publishing and archive.

WEB-09 FAQs

- Published central FAQs can be selected for Programs or Montessori Way and included in search.

WEB-10 Testimonials

- Text testimonials only: named or anonymised attribution, relationship, optional consented photo, related program, feature/order and placement controls. No ratings, video or feedback forms.

WEB-11 Team

- Name, photo, designation, short profile, qualifications, experience, specialisation/responsibility, order, featured/active status and consent. No private contacts or identity documents.

WEB-12 Documents

- Published PDFs with title, description, fixed module classification, version/replacement support, visibility and usage locations.

WEB-13 Contact and location

- Address, landmark, hours, embedded Google Map, directions, phone, WhatsApp and enquiry form, all centrally configurable.

WEB-14 Search

- Global search covers only published pages, Programs, Experiences, Events/Announcements, Blog/News, FAQs and documents; results show title, summary and type.

WEB-15 Error and maintenance

- Branded 404 with useful links. Super Admin can enable maintenance mode, provide message/reopening time and contacts while retaining admin access.

5. Content Management Requirements

CMS-01 Dashboard

- Role-aware summaries for awaiting review, recent/scheduled content, enquiries, follow-ups, media failures, recycle-bin deadlines and activity; provide permitted quick actions.

CMS-02 Lifecycle

- Content statuses: Draft → In Review → Approved/Scheduled → Published → Unpublished/Archived → Recycle Bin. Only authorised roles may transition each state.

CMS-03 Review comments

- Publisher/Super Admin can comment; rejection requires a comment; editors can revise and resubmit; comments and transitions stay in history.

CMS-04 Responsive preview

- Secure preview for mobile, tablet and desktop. Editors see their own drafts; Publisher/Super Admin see submitted content. Preview displays a Draft Preview indicator and is not publicly indexable.

CMS-05 Scheduling

- Publisher/Super Admin schedule any publishable content in Asia/Kolkata, edit/cancel schedules, and receive an in-app failure alert. Editors may suggest dates only.

CMS-06 Version history and rollback

- Persist meaningful revisions, author, timestamp and action. Authorised roles can compare and restore an earlier version without erasing the audit trail.

CMS-07 Autosave

- Editing screens autosave drafts, show saving/last-saved status and warn before leaving with unsaved changes. Concurrent edit locking is excluded.

CMS-08 Validation

- Publishing checks required fields, alt text, child-image consent, links, SEO metadata, unique slug, logical schedules/expiry, preview acknowledgement and media readiness. Errors block; authorised users may acknowledge warnings.

CMS-09 Duplication

- Pages, Programs, Experiences, Events, FAQs and articles can be copied into new drafts. Slug is cleared/regenerated; dates, approvals, analytics and enquiries are not copied; media reuse is optional.

CMS-10 Lists and filters

- Search/filter by title, type, status, owner, fixed classification, dates and featured flag. Phase 1 permits ordinary individual actions; advanced bulk workflow actions are excluded.

CMS-11 Recycle Bin

- Deleted content is recoverable during a configured retention window; dashboard warns near permanent deletion. Super Admin may restore or permanently delete according to permissions.

CMS-12 Manual related content

- Administrators select and order related Programs, Experiences, Events, articles and FAQs. Unpublished/archived items disappear automatically.

CMS-13 Form configuration

- Super Admin enables forms, selects approved fields, required/optional state, headings, intro/success/consent text, enquiry type/program and in-app recipients. No unrestricted or sensitive custom fields.

CMS-14 Brand settings

- Super Admin manages logos, favicon, approved colours/fonts, default sharing image, buttons and header/footer appearance within a controlled accessible design system.

CMS-15 Activity log

- Record sign-ins, account/role/status changes, publishing, scheduling, review, rollback, imports/exports where applicable, deletions, media actions and system-setting changes with actor and timestamp.

6. Media Management Requirements

MED-01 Organisation

- Search filename/title/keyword; filter by type, fixed classification/folder, uploader/date, used/unused status, child consent, dimensions and size; support tagging/folder organisation local to the module.

MED-02 Usage

- Show every content item using an asset. In-use media cannot be permanently deleted until replaced or removed.

MED-03 Images

- Validate safe supported types, integrity, size and dimensions; correct orientation; retain original; create thumbnails/responsive sizes; compress and convert suitable images to WebP.

MED-04 Video

- Accept standard validated uploads within agreed limits. Automatic transcoding and thumbnail generation are excluded from Phase 1.

MED-05 Processing status

- Display Uploading, Processing, Ready or Failed; allow retry; required media must be Ready before publishing.

MED-06 Consent

- Identifiable child images require recorded consent status and validation before public use. Revoked/absent consent prevents publication.

7. Enquiry Management Requirements

ENQ-01 Capture

- Approved public forms create an enquiry with source, type, related program, submitted fields, consent version, timestamps and status. Apply server-side validation and Firebase rules.

ENQ-02 Workflow

- Support New, Assigned, In Progress, Follow-up Due, Closed and Spam (or equivalent approved labels), with history.

ENQ-03 Assignment

- Super Admin assigns/reassigns enquiries. Assigned users filter their work and receive an in-app notification; changes are logged.

ENQ-04 Follow-ups

- Record date/time, method (Phone, WhatsApp, Email, In Person, Other), outcome/internal notes, next date, responsible administrator and completion.

ENQ-05 Reminders

- Upcoming and overdue follow-ups appear on the admin dashboard. No automatic external messages are sent.

ENQ-06 Export

- Super Admin exports CSV or Excel filtered by date, type, program and status. Exclude internal notes by default and log every export.

ENQ-07 Deletion

- No automatic retention deletion. Only Super Admin may permanently delete after confirmation; deletion is logged. Final privacy policy may introduce a future retention rule through change control.

8. SEO, Analytics and Integrations

SEO-01 Metadata

- Each public item supports SEO title, meta description, unique readable slug, canonical URL, indexing control and social title/description/image with length/duplicate warnings.

SEO-02 Discovery

- Generate XML sitemap and robots.txt; include only published/indexable content. Provide schema.org data for school, Programs, Events, Articles, FAQ and breadcrumbs.

SEO-03 Preview

- Show Google-style and social-sharing previews before publishing.

SEO-04 Redirects

- Publisher/Super Admin manage 301/302 redirects; suggest redirects after published slug changes; prevent loops/conflicts; detect broken internal links and locations; count redirect visits.

INT-01 Social

- Centrally manage profile links, sharing buttons and WhatsApp sharing. Optional approved Instagram/YouTube showcases must be lazy/performance optimised and respect consent. No automatic posting.

INT-02 Analytics

- Integrate approved analytics only after applicable cookie consent. Provide basic dashboard summaries; configure Search Console for production.

INT-03 Maps

- Use an editable Google Maps URL/embed and Get Directions action without exposing unrestricted configuration to ordinary users.

9. Legal, Privacy and Consent

LEG-01 Policy pages

- Editable Privacy Policy, Terms and Conditions, Cookie Policy, Disclaimer and Child Image and Media Policy. Show last updated date; retain version history; publishing restricted to Publisher/Super Admin.

LEG-02 Cookie choices

- Visitors can Accept All, Reject Optional or choose preferences. Essential cookies remain active. Cookie Settings allows later change.

LEG-03 Consent record

- Record cookie choice and policy version with minimal data. Analytics/social tracking must not load before required consent.

LEG-04 Data minimisation

- Phase 1 forms use approved fields only and do not solicit sensitive child, medical, identity-document or payment data.

LEG-05 Confidentiality

- Internal enquiry notes, account data and audit records are never public and are accessible only to authorised roles.

10. Technical Architecture

The following is the approved implementation baseline unless the engineering design review documents and approves an equivalent alternative.

Layer	Baseline responsibility
Web application	Responsive public site and admin portal, compatible with Firebase Hosting/App Hosting; support server-rendered or pre-rendered public pages where required for SEO.
Authentication	Firebase Authentication, email/password only in Phase 1.
Database	Cloud Firestore for structured content, users, workflows, settings, enquiries, logs and metadata.
Files	Firebase/Google Cloud Storage for originals and renditions, protected by access rules.
Server operations	Trusted Cloud Functions/server environment for user administration, claims, publishing schedules, exports, media processing and privileged deletions.
Scheduling	Server-side scheduled jobs using Asia/Kolkata-aware timestamps; store canonical timestamps consistently and render in configured timezone.
Security enforcement	Firebase Security Rules plus server-side authorisation; UI hiding alone is never considered access control.

Layer	Baseline responsibility
Observability	Structured error logging, publishing/media failure states and admin-visible operational alerts.
Environments	Separate development/testing and production configuration; production secrets must not be committed to source control.

10.1 Suggested logical collections

Collection/domain	Examples
users / roles	Profile, role, status, mustChangePassword, audit fields
content	Pages and structured module records; status, owner, slug, SEO, schedule, versions
media	Original/renditions, metadata, consent, usage, processing state
enquiries	Submission, assignment, status, follow-ups, internal notes
settings	Brand, contact, navigation, footer, cookie, forms, maintenance
auditLogs / notifications	Immutable administrative events and in-app notifications
redirects	Source, destination, type, active status, hit count

11. Non-Functional Requirements

ID	Area	Acceptance target
NFR-01	Responsive design	No horizontal overflow; complete key journeys at 360 px mobile width, tablet and desktop breakpoints.
NFR-02	Accessibility	Target WCAG 2.2 Level AA: keyboard access, visible focus, semantic headings/landmarks, labels/errors, alt text, sufficient contrast and reduced-motion consideration.
NFR-03	Performance	Production pages should target Lighthouse ≥90 for Performance, Accessibility, Best Practices and SEO on representative templates, subject to documented third-party limitations.
NFR-04	Core Web Vitals	Target good thresholds at launch: LCP ≤2.5 s, INP ≤200 ms and CLS ≤0.1 on representative mobile testing.
NFR-05	Security	Least privilege, server validation, protected secrets, secure file rules, no plaintext passwords, authorisation tests for every role and sensitive operation.
NFR-06	Reliability	Failed scheduled publishing/media work remains visible and retryable without silent data loss.
NFR-07	Auditability	Administrative actions carry actor, timestamp, target, action and outcome; logs are not editable by ordinary

ID	Area	Acceptance target
		administrators.
NFR-08	Compatibility	Latest stable Chrome, Edge, Safari and Firefox; current common Android/iOS browsers.
NFR-09	SEO quality	Indexable published pages have canonical metadata, sitemap presence and valid structured data; drafts/previews are noindex and inaccessible publicly.
NFR-10	Backup/recovery	Production launch requires documented Firestore/Storage backup, restore ownership, cadence and tested recovery procedure.
NFR-11	Privacy	Collect only necessary data; show consent language; restrict exports and internal notes; complete privacy/retention review before launch.
NFR-12	Maintainability	Use reusable typed schemas/components, environment separation, automated checks and documented deployment/rollback steps.

12. Key Workflows

12.1 Content publishing

- Editor creates or updates a draft; autosave records progress.
- Editor previews responsive layouts, completes validation and submits for review.
- Publisher/Super Admin reviews, comments and either rejects with a mandatory reason or approves.
- Authorised reviewer publishes immediately or schedules publication.
- System validates again, publishes atomically, updates sitemap/search and records the event.
- Failure leaves the previous public version intact where feasible, raises an in-app alert and supports retry.

12.2 User onboarding

- Super Admin creates the account with a temporary password.
- User signs in with the same email and temporary password.
- System requires password change before normal portal use.
- User edits permitted profile fields; role/status remain controlled.

12.3 Enquiry follow-up

- Visitor submits an approved form and sees a configurable success message.
- System creates a New enquiry and alerts configured administrators in-app.
- Super Admin assigns it; assigned administrator records interactions and next follow-up.
- Dashboard shows upcoming/overdue follow-ups until completed or enquiry closed.

13. Acceptance Criteria by Release Capability

Capability	Release acceptance
Public site	All approved modules are responsive, navigable and populated with production-approved content; unpublished content is never exposed.

Capability	Release acceptance
Administration	Role matrix is enforced server-side; account creation, forced password change, suspension and profile editing work as specified.
Publishing	Draft/review/comment/approval/preview/schedule/publish/archive/version/rollback paths pass end-to-end tests.
Media	Image validation, renditions, consent blocking, usage display and in-use deletion protection pass.
Enquiries	Capture, assignment, follow-up, reminders, filters, export and Super Admin deletion pass with audit entries.
Search/SEO	Published content appears; drafts do not. Sitemap, redirects, canonical data and structured data validate.
Legal/consent	Policy pages, consent choices and deferred analytics behaviour pass in supported browsers.
Quality	Accessibility, browser, mobile, security-rule, performance, backup/restore and production smoke checks meet documented targets.
Operations	Deployment, rollback, ownership, monitoring and incident contacts are documented and handed over.

14. Testing and Launch Checklist

- ☐ Final content, branding, contact details and consent evidence approved
- ☐ Role/permission matrix and Firebase Security Rules tested with positive and negative cases
- ☐ Authentication, temporary-password and account-suspension flows tested
- ☐ Every content module and status transition tested
- ☐ Scheduled publishing tested using Asia/Kolkata times and failure handling
- ☐ Media limits, image optimisation, consent and deletion safeguards tested
- ☐ Enquiry consent, assignment, follow-up, export and deletion tested
- ☐ Policy pages and cookie choices reviewed by the responsible stakeholder
- ☐ Accessibility keyboard/screen-reader spot checks and automated audit completed
- ☐ Lighthouse and Core Web Vitals tested on representative pages
- ☐ Chrome, Edge, Safari, Firefox, Android and iOS smoke tests completed
- ☐ 404, broken-link detection, redirects and maintenance mode tested
- ☐ Domain, SSL, analytics consent, Search Console, sitemap and robots configured
- ☐ Production backups, restore test, monitoring and rollback procedure approved
- ☐ No test users, sample enquiries, secrets, debug flags or placeholder copy remain
- ☐ Final stakeholder user-acceptance sign-off recorded

15. Phase 2 Backlog

Area	Deferred capability
Admissions	Full application workflow, document upload, applicant tracking and payments
Language	Tamil content authoring, translation workflow and public language switcher
Authentication/communication	Google Sign-In, invitation emails, email/WhatsApp automation
Forms/security	Advanced bot/spam detection, rate limiting, duplicate prevention and conditional CAPTCHA

Area	Deferred capability
Content operations	CSV/Excel content import, advanced bulk workflows and concurrent edit locking
Taxonomy/discovery	Shared categories/tags and automatic related-content recommendations
Media	Automatic video transcoding/thumbnails and advanced duplicate detection
Social/feedback	Automatic social posting, ratings, video testimonials and parent feedback forms

16. Risks, Decisions and Open Items

Item	Required decision/mitigation	Owner / timing
Scope size	Estimate module-by-module and protect exclusions through change control.	Product + delivery team / before sprint plan
Media/video cost	Confirm maximum formats, duration, size, storage and delivery approach.	Product + engineering / before media build
Privacy retention	Phase 1 has manual enquiry deletion; obtain final policy/legal review.	Business owner / before production
Backup recovery	Define cadence, retention, restore test and accountable owner.	Engineering / before production
Analytics/social embeds	Load only after applicable consent and measure performance impact.	Engineering + product / before launch
Content readiness	Prepare approved copy, images, alt text and consent evidence early.	2D Montessori / during build
Technical design	Confirm framework, environment structure, indexing/search method and scheduled-job design.	Engineering / design review

17. Change Control and Sign-off

This version is the Phase 1 baseline. A proposed change should state the business reason, affected requirements, design/data/security impact, delivery estimate and acceptance-test changes. Implementation begins only after the authorised product owner approves the change.

Approval	Name	Signature / confirmation	Date
Product Owner / Business	Sivakesh Raman		
Technical Lead			
Quality / UAT			

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